

Analytical problem solving Activity

1. Gradient Magnitude and Gradient direction

Given: $G_x = 24$, $G_y = 18$

Gradient Magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

$$G = \sqrt{24^2 + 18^2} = \sqrt{900} = 30$$

Gradient Direction.

$$\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$$

$$= \tan^{-1}\left(\frac{18}{24}\right)$$

$$= \tan^{-1}(0.75)$$

$$\theta = 36.87^\circ$$

$\therefore$ Gradient Magnitude = 30

Gradient Direction = 36.87°

2. Non-Maximum Suppression

Pixel	P_1	P_2	P_3	P_4	P_5
Magnitude	25	50	90	60	30

$P_1 = 25$ - suppressed.

$P_2 = 50$ - suppressed

because $P_3 = 90$ is larger.

$P_3 = 90$ - Retained because it is greater than $P_2 = 50$ & $P_4 = 60$.

$P_4 = 60 \rightarrow$ suppressed because $P_3 = 90$ is larger.

$P_5 = 30 \rightarrow$ suppressed

After NMS:

Pixel	P_1	P_2	P_3	P_4	P_5
Result	0	0	90	0	0

$\therefore P_3 = 90$
remaining edge pixel.

3. Double Thresholding

High threshold = 100
low threshold = 50

Pixel	Magnitude
P ₁	135 > 100
P ₂	90 < 100
P ₃	45 < 50
P ₄	120 > 100
P ₅	70 < 100
P ₆	20 < 50

Classification

strong edge
weak edge
non edge
strong edge
weak edge
non edge

Strong edges: P₁, P₄
weak edges: P₂, P₅
non edges: P₃, P₆

4. Edge Tracking by Hysteresis

Pixel	Magnitude	Classification
P ₁	130	strong
P ₂	95	weak
P ₃	40	non edge
P ₄	115	strong
P ₅	65	weak
P ₆	25	non edge
P ₇	105	strong

1. Strong edges.

magnitude ≥ 100

They are: P₁, P₄, P₇

2. weak edges: magnitude $50 \leq x < 100$

They are: P₂, P₅

3. Non edge:

magnitude < 50

P_3, P_6

Final edge pixels after Hysteresis

P_2 is connected to $P_1 \rightarrow P_2$ becomes a valid edge

P_5 is not connected to any strong edge $\rightarrow P_5$ is removed.

$\therefore$ The connected pixels are P_1, P_2, P_4 & P_7 .

4. Hough circle transform

1. Given

$$P_1 = (5, 4) \quad P_2 = (7, 6)$$

$$\text{Candidate centers } C_1 = (2, 4) \quad C_2 = (4, 3)$$

$$\text{for } C_1: r_1 = \sqrt{(5-2)^2 + (4-4)^2} = \sqrt{3^2} = 3$$

$$r_2 = \sqrt{(7-2)^2 + (6-4)^2} = \sqrt{5^2 + 2^2} = \sqrt{29} = 5.39$$

$$\text{for } C_2: r_1 = \sqrt{(5-4)^2 + (4-3)^2} = \sqrt{1+1} = \sqrt{2} = 1.41$$

$$r_2 = \sqrt{(7-4)^2 + (6-3)^2} = \sqrt{3^2 + 3^2} = \sqrt{18} = 4.24$$

$\therefore$ The centers do not produce matching radius values for both edge points. Therefore, neither C_1 nor C_2 is a valid common center for P_1 & P_2

2. Given,

$$P = (8, 6) \quad C = (4, 3)$$

$$\begin{aligned} r &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(8-4)^2 + (6-3)^2} \\ &= \sqrt{4^2 + 3^2} \\ &= \sqrt{16+9} \\ &= \sqrt{25} = 5 \end{aligned}$$

3. Given

$$P = (8, 8)$$

$$C = (5, 4)$$

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(8 - 5)^2 + (8 - 4)^2}$$

$$= \sqrt{3^2 + 4^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$r = 5$$

$\therefore$ the radius = 5